RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

ESX Cluster-Store (DKVS) Loop & vpxd Crash Troubleshooting

Diagnosing and remediating the ESX cluster-store (DKVS) reconfigure loop behind vpxd crashes (KB 385346, fixed in 9.1) — is_in_alarm vs the hot loop, the half-joined replica, and the monitor.

v1.0 VMware Cloud Foundation 9

INTERNAL

ESX Cluster-Store (DKVS) Reconfigure-Loop & vpxd Crash Troubleshooting

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Platform:** VMware Cloud Foundation 9.0.1

> **When to use.** vCenter's `vpxd` service crashes and auto-restarts, and/or `vpxd.log` repeatedly logs cluster-store replica errors. This runbook is the reusable how-to; the full incident analysis is in [[RCA-vpxd-Failure-Cluster-Store-InvalidState-2026-07-27]].

1. Symptoms

- `systemctl status vpxd` → **failed**, `Main PID … code=killed, signal=TERM`; `ExecStartPost/StopPost` `status=4/NOPERMISSION`.
- `vpxd.log` (recurring): `DistEsxClusterStoreManager … RECONFIGURE … InvalidState calling 'runReplica'`, `Failed to add replica to cluster store; [HostSystem:host-NN, esxiNN]`, and/or `Failed to create user … UserAlreadyExists calling 'createUser'`.
- vCenter usually **recovers on its own** (vMon restarts vpxd).

2. Diagnose (read-only)

On the VCSA (`ssh root@vcenter` → `shell`):

```
systemctl status vpxd           # NOTE: vpxd is vMon-managed — systemd may show "failed/failed" and stale timestamps; NOT authoritative
vmon-cli -s vpxd                # authoritative health (want HEALTHY)
pidof vpxd                      # a CHANGE between checks = vpxd was recycled (a new incident)
grep -c "InvalidState calling"  /var/log/vmware/vpxd/vpxd.log      # loop count (rising = active)
grep    "InvalidState calling"  /var/log/vmware/vpxd/vpxd.log | tail -3    # which host + how recent
df -h ; date -u                 # cheap ruleouts (full partition / clock skew)
```

Cluster-store state — run on the primary ESXi host **and** the suspect host:

```
/usr/lib/vmware/clusterAgent/bin/clusterAdmin cluster status
```

Read: `is_in_alarm` (the escalation trigger), `members`, and each member's **`api_address`** — a member with a `peer_address` (:2380) but **blank** **`api_address`** (:2379) is a **half-joined replica** (in the replication layer, not serving) — the footprint of a failed `runReplica`.

`is_in_alarm: false` **does NOT mean healthy.** The store *quorum* can be fine (primary serving, `reach`-equivalent OK) while the **reconfigure machinery loops hot**. Always check **both** `clusterAdmin cluster status` (quorum/alarm) **and** the `vpxd.log` loop count/recency.

3. Root Cause

The ESX **cluster store (DKVS — Distributed Key-Value Store**, etcd across hosts on :2379/:2380) has a replica stuck reconfiguring — a host can't complete joining as a serving member. Broadcom **KB 385346**: known issue on **ESX 9.x, permanently fixed in vSphere 9.1** (ESXi 8.0 U3g on the 8.x line). A reconfigure during the loop can trip vMon into cycling vpxd.

4. Remediation

4.1 Confirm state first (read-only)

`clusterAdmin cluster status` → if **`is_in_alarm: false`**, the KB's emergency workaround (§4.3) is **NOT indicated** — the store is serving. If the loop is active but not in alarm, monitor (§5) and plan the upgrade.

4.2 Permanent fix (recommended)

Upgrade to vSphere 9.1 — contains the fix; the durable resolution for the incomplete-replica/loop condition.

4.3 Workaround — disable DKVS (KB 385346) — only if `is_in_alarm: true`

Mutating — restarts vpxd (~2 min vCenter unavailable), needs a fresh backup, can force host reconnects. Only run if the store is genuinely in alarm.

```
 # On vCenter:
/usr/lib/vmware-vpx/py/xmlcfg.py -f /etc/vmware-vpx/vpxd.cfg set vpxd/clusterStore/globalDisable true
vmon-cli -r vpxd
# On EACH ESXi host:
/etc/init.d/clusterAgent stop
configstorecli files datafile delete -c esx -k cluster_agent_data
configstorecli files datadir  delete -c esx -k cluster_agent_data
```

Broadcom also ships a `dkvs-cleanup.py` helper (disable + host cleanup in one pass). Verify: the `globalDisable` key reads `true`.

5. Monitoring

Use the **cluster-store loop monitor** (`06-Troubleshooting/clusterstore-monitor/`) — a read-only checker (scheduled every 30 min) that watches `is_in_alarm` , store membership/half-join, the `vpxd.log` loop count, and the **vpxd PID** (a change = a new recycle). Status: **CRIT** on alarm or PID change; **WARN** = loop active but not in alarm; **OK** = quiesced.

6. Capture-at-Failure (to close the loop)

If vpxd cycles again, grab `vpxd.log` **at the moment of failure before it rotates** — those lines convert the probable-cause link from *inferred* to *confirmed* (see the RCA §5).

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Document End